

HOW POPULATIONS EVOLVE

Module map

- Module 13: How Populations Evolve
- Connected lab: lab-13_how-populations-evolve.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Allele frequencies as
the unit of change

Sources of
variation

Hardy-Weinberg as
a baseline

Module 13

Five
mechanisms
of
microevolution

Genetic
drift
and
selection

Lab 13 key-
populations-
evolve.md

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Learning objectives

- Define microevolution using allele frequencies.
- Compare the five mechanisms that change allele frequencies.
- Distinguish bottleneck and founder effects.
- Interpret directional, stabilizing, and disruptive selection.
- Use Hardy-Weinberg as a non-evolving baseline.

OBJECTIVE 1

Define microevolution using allele frequencies.

OBJECTIVE 2

Compare the five mechanisms that change allele frequencies.

OBJECTIVE 3

Distinguish bottleneck and founder effects.

OBJECTIVE 4

Interpret directional, stabilizing, and disruptive selection.

OBJECTIVE 5

Use Hardy-Weinberg as a non-evolving baseline.

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Topic sequence

- Allele frequencies as the unit of change: Microevolution is change in allele frequencies across generations.
- Sources of variation: Mutation creates new variation, while recombination reshuffles existing variation.
- Five mechanisms of microevolution: Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.
- Genetic drift and selection: Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.
- Hardy-Weinberg as a baseline: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

TOPIC 1

Allele frequencies as the unit of change

TOPIC 2

Sources of variation

TOPIC 3

Five mechanisms of microevolution

TOPIC 4

Genetic drift and selection

TOPIC 5

Hardy-Weinberg as a baseline

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Module 13: How Populations Evolve Evidence Map

- Module focus: How Populations Evolve
- Central claim: How Populations Evolve explains allele-frequency change through drift, gene flow, selection, and baseline models.
- Key nodes to connect: Allele frequencies as the unit of change, Sources of variation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-13_how-populations-evolve.md supplies an evidence surface for the map.

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Module 13: How Populations Evolve Reasoning Sequence

- Module focus: How Populations Evolve
- Trace the model: Allele frequencies as the unit of change -> Sources of variation -> Five mechanisms of microevolution -> Genetic drift and selection
- Inputs become outputs: Allele frequencies as the unit of change, Sources of variation -> Define microevolution using allele frequencies., Compare the five mechanisms that change allele frequencies.
- Feedback check: lab-13_how-populations-evolve.md evidence checks whether students can explain five mechanisms of microevolution.

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Terms as evidence handles

- Allele frequency: Proportion of a particular allele in a population.
- Microevolution: Allele-frequency change across generations.
- Mutation: DNA sequence change that can create new alleles.
- Gene flow: Movement of alleles between populations.
- Genetic drift: Random allele-frequency change, strongest in small populations.
- Bottleneck effect: Drift after a sharp population reduction.

ALLELE FREQUENCY

Proportion of a particular allele in a population.

MICROEVOLUTION

Allele-frequency change across generations.

MUTATION

DNA sequence change that can create new alleles.

GENE FLOW

Movement of alleles between populations.

GENETIC DRIFT

Random allele-frequency change, strongest in small populations.

BOTTLENECK EFFECT

Drift after a sharp population reduction.

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Lab connection

- Lab file: lab-13_how-populations-evolve.md
- Lab evidence should test or illustrate:
Five mechanisms of microevolution
- Students should leave with a
concrete observation, comparison, or
model tied to the module claim.

QUESTION

Allele frequencies as
the unit of change

EVIDENCE

lab-13_how-
populations-
evolve.md

CLAIM

Define microevolution
using allele
frequencies.

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Contrast check

- Do not stop at: Microevolution is change in allele frequencies across generations.
- Stronger explanation: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Microevolution is change in allele frequencies across generations.

DEEPER

Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

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Module 13: How Populations Evolve Retrieval and Lab Check

- Module focus: How Populations Evolve
- Retrieval prompt: What is an allele frequency?
- Answer check: Define microevolution using allele frequencies.
- Required terms: Allele frequency, Microevolution, Mutation
- Lab connection: lab-13_how-populations-evolve.md; lab-13_how-populations-evolve.md supplies evidence for population evolution evidence from frequency changes.

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Practice quiz bridge

- 1. Which statement best defines Allele frequency in Module 13?
- 2. Which learning goal best supports the topic "Sources of variation"?
- 3. How should lab-13_how-populations-evolve.md support Module 13?
- 4. Which retrieval move best prepares a student for How Populations Evolve?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

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Synthesis and exit ticket

- Exit claim: explain allele frequencies as the unit of change using evidence from lab-13_how-populations-evolve.md.
- Must include: Allele frequency, Microevolution, Mutation.
- Revision prompt: what evidence would change your explanation?

CLAIM

Allele frequencies as the unit of change

EVIDENCE

lab-13_how-populations-evolve.md

REVISION

What would change your mind?